

LINGUISTIC CHARACTERISTICS RELATED TO SUICIDALITY USING AN EXISTING
DATA ARCHIVE OF REDDIT POSTS

by

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This thesis was prepared under the direction of the candidate's thesis advisor, Dr. Kevin Lanning, and has been approved by the members of their supervisory committee. It was submitted to the faculty of The Wilkes Honors College and was accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Liberal Arts and Sciences.

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ABSTRACT

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The language that we use often expresses our mental characteristics, including any psychological difficulties we may be experiencing. Thus, we may be able to detect certain psychopathological phenomena, such as suicidal ideation, from language; this has potential practical applications to detection and treatment. Review of theoretical and empirical literature suggests that certain features of language style and content are associated with the presence of suicidality. The present study investigates the linguistic characteristics of Reddit posts made by users that either exhibited or did not exhibit a signal of suicidality by either posting to or not posting to r/SuicideWatch, respectively. Our results indicated that these two groups of Reddit users differed on a variety of language dimensions, as measured by the Linguistic Inquiry and Word Count (LIWC), and that the predictive logistic regression models we built were able to accurately differentiate between these two groups a majority of the time.

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Introduction

Suicide is a widespread public health crisis that will require continued investigation and effective interventions to be successfully managed. The World Health Organization (2019) reports that approximately 700,000 people died by suicide in 2019, accounting for 1.3% of all deaths around the world. Despite the pervasiveness of suicide and its widespread negative consequences, our ability to predict suicide is poor, with over half a century of reviewed research on the subject yielding no definitive conclusions on risk factors (Franklin et al., 2017). A variety of theories that attempt to explain the phenomenon of suicide have been advanced, with many being partly inaccurate. The low base rate at which suicide occurs is likely somewhat responsible for this, as it is statistically difficult to predict phenomena that are uncommon (Mitchell et al., 2021). Additionally, suicide is probably best understood as the result of many different factors interacting, further complicating prediction (Leenaars, 1996).

Franklin et al. (2017) concluded, from their meta-analysis of research into risk factors for suicide, that risk factors that currently guide clinical practice are not good at predicting the occurrence of suicide. Risk factors include things like “stressful life events”, “depression”, and “prior suicide attempt”, which seem like they would be associated with suicidal thoughts and behaviors to an extent that would allow them to be used as identifiers of suicidal people. However, research does not support this idea; the risk factors that Franklin et al. (2017) reviewed were unable to predict suicidal thoughts and behaviors accurately or strongly. Additionally, our ability to treat suicidality prospectively has not improved in recent years, with many treatments being empirically tested yet only a small number showing reduction of suicidal thoughts and behaviors in their participants. These are alarming conclusions for any clinician, scientist, or

policymaker interested in reducing the frequency of suicide. More research is certainly needed to clarify relationships between suicidality and other thoughts, emotions, behaviors, and life events.

Before continuing with our discussion of suicide, however, we define the terminology that we will use to ensure clarity. Silverman (2006) argues in favor of a universal vocabulary for discussing suicidality and attempts to define one using the terms that research on suicide currently uses. He states that a specific set of terms about suicide is critical to understanding it and enabling effective communication about it, the lack of which impedes our ability to learn about and deal with suicide. After reviewing literature on the topic, he comes to the conclusion that *suicide* is a death in which the deceased person was aware of the threat of dying and intentionally inflicted their death upon themselves. This definition leaves room for what Silverman refers to as *indirect* or *passive suicide*, in which a person may not have taken explicit action toward causing their own death but nevertheless was aware of and accepted the threat of dying. An example of this might be a person who chooses to engage in risky behavior, such as driving while intoxicated, because they feel apathetic to the threat of death. *Suicidal behavior* is distinguished from such risk-taking behavior by the person's intent to die and the objective potential lethality of the action taken, while a *suicide attempt* is a concrete action taken by a person that includes both a high probability of death and the intention to die.

Silverman also states that the terms *suicidality* and *being suicidal* are extremely loosely defined and are used to describe a wide range of behaviors, from thinking about to planning to attempting suicide. As such, these terms are not often useful in nuanced discussion of specific factors of suicide; however, we choose to use this term moving forward in the present study as our work focuses on identifying general characteristics of suicidality and is mostly exploratory in nature.

Theories of Suicide

As we discuss previously, over half a century of research on suicide has failed to come up with a definitive explanation for the phenomenon, and thus there is no scholarly consensus on the cause(s) of suicide (Franklin et al., 2017). A variety of discrete theories have been advanced to attempt to explain the phenomenon of suicide, and they take a multitude of perspectives, including biological, sociocultural, and psychological; however, none account for all of the components of suicide. A few that we believe to be especially explanatory approach suicide from the perspective of psychology and focus on the themes of escape, mental pain, and narrowing of consciousness. Additionally, we focus on psychological theories of suicide because, as Leenaars (1996) points out, suicide is best understood through the lens of psychology as the motivation for suicide takes place in the mind.

The first key point about suicide is that, as Shneidman (1998) puts it, “Suicide is never born out of exaltation or joy” (p. 245). There must be some negative phenomenon pushing the suicidal person towards the decision to end their own life. Shneidman theorized that this phenomenon is the subjective experience of mental pain, or *psychache*, as he terms it. He provides a quote that influenced his thinking on the subject from his mentor, Henry Murray: “What is suicide but an effort to stop the unbearable flow of negative affects?” (Shneidman, 1998, p. 245). He also notes that, during his tenure as a clinical psychologist, most cases of suicide that he witnessed appeared to stem from pain (particularly pain which is psychological in nature) and a desire to end it. For suicidal individuals, he posits, the pain has crossed the threshold into unbearability; it is no longer tolerable, and so they come to the conclusion that the only way to escape this pain is to die. Shneidman argues that psychache is only ever borne out of

negative life circumstances, such as having a mental disorder or experiencing strong negative emotion, which to a suicidal person qualify as strong motivators to end one's life.

Shneidman also argues that suicide is mainly a mental act, as opposed to other conceptions of suicide that propose it is interpersonal or sociological in nature. He states:

I believe that we ought to concentrate on the mentalistic aspects of suicide, to stick with the phenomenology of suicidal acts, to focus on the suicidal drama in the life of the mind. Obviously, there is no mind without the neurological activity of a brain, but this physiological fact does not preclude the formulation that the mind has a mind of its own, and that the main business of the mind is to mind its own business. (Shneidman, 1998, p. 245)

To him, understanding suicide must begin with understanding the mental motivations for suicide. He splits the theoretical literature on suicide from the twentieth century into four groups of motivators, which he calls *fight* (suicide as an expression of anger or aggression), *flight* (suicide as an expression of a desire to escape or find relief), *fright* (suicide as a response to negative emotion or pain), and *freight* (suicide as the result of unfortunate environmental or life conditions, such as abuse or a family history of mental illness). In his clinical experience, the feelings of most suicidal individuals that he has interacted with appeared to stem from the motivator of flight. More specifically, he believes that they came from a desire to escape psychache; thus, searching for expression of a desire to escape from one's mental pain may be a useful way to detect the presence of suicidality in an individual, especially in language as the present study attempts to do.

Shneidman (1993) also describes the state of mind that the suicidal person tends to be in directly before their suicide as a state of *narrowed consciousness*, in which they become resistant

to any information from either internal or external sources that may change their mind about their decision to commit suicide. Narrowing of consciousness results in the person feeling as though the only way to resolve their problems is suicide, as everything begins to feel overwhelming and unsolvable. This mental state sounds similar to absolutist (or “black-and-white”) thinking and cognitive deconstruction, the latter of which has been addressed by other theorists, including Baumeister (1990) who defines *cognitive deconstruction* as a mental transformation to lower levels of awareness, thinking, and consciousness. Cognitive deconstruction acts as a sort of defense mechanism for the person experiencing extreme mental pain because it lowers their ability to experience as a fully conscious and autonomous being and thus reduces the amount of pain they may feel. Cognitive deconstruction is also associated with extreme negative or neutral feelings and with apathy, which tend to be anecdotally associated with suicidality as many suicides tend to be perceived as unexpected (e.g., McHugh et al. (2019) found that about 60% of people that commit suicide do not disclose their suicidal ideation before their deaths).

Baumeister’s idea of cognitive deconstruction updates an older escape theory of suicide (Baechler, 1979, as cited in Baumeister, 1990). The model described by Baumeister contains six steps, where a “positive” outcome (i.e., the situation outlined in the step actually occurs) at each step results in the person approaching a suicide attempt: (1) the person has a negative experience that leads to some expectation of theirs remaining unmet, leading to disappointment; (2) the person internalizes and personalizes the cause of the disappointment, which leads to them to feeling as though the situation is a reflection upon themselves; (3) the person becomes extremely and unpleasantly aware of themselves because they compare their current circumstances to their expectation; (4) this comparison leads to negative feelings, such as self-perceptions of

incompetence or inadequacy; (5) the person responds to these negative feelings by entering a state of cognitive deconstruction in an attempt to escape them, although this does not entirely work; (6) as a side effect of the lack of inhibition from the state of cognitive deconstruction and additionally as a response to the desire to escape, suicide may arise to the person as a feasible option and so they may attempt suicide as a coping action.

While each individual instance of suicide is influenced by factors unique to the person that committed suicide, there are general patterns of experience that tend to be related to suicide, as we describe above. If we uncover what these patterns are, we can investigate how they tend to be expressed through language and use this information to identify the likely presence of suicidality in the author of a piece of text. In other words, these theories of suicide can inform our search for the linguistic characteristics of suicidality, as we can search for the ways that the suicidal mental state manifests in language.

Suicide Research Using Natural Language Processing

Shneidman (1993) was also interested in the ability to understand suicide through the language of suicide notes; however, he points out that, because of the state of cognitive deconstruction experienced by the suicidal individual prior to committing suicide, suicide notes often fail to inform the reader about the motivation behind the suicide. He argues that the state of mind that a person is in prior to their death by suicide prevents them from accurately conveying their thoughts or reasoning for committing suicide:

In order for a person to kill himself he has to be in a special state of mind, a state of relatively fixed purpose [...] and relative close-mindedness. It is a psychological state which, while it permits suicide to occur, obviously militates against insight or good communication. In other words, that special state of mind necessary to perform the

suicidal deed is one which is essentially incompatible with an insightful recitation of what was going on in one's mind that led to the act itself. (Shneidman, 1993, p. 98)

The idea that a suicide note would be able to provide insight into why a person chose to commit suicide seems intuitive, but extracting meaning from these documents is often difficult due to the state of mind that genuinely suicidal individuals are usually in before committing suicide.

It is possible, though, that this mental state and other characteristics of suicidality could be detected within text produced by a suicidal individual. The language that a person uses expresses and reflects their psychological state both explicitly and implicitly, making it a valuable source of data for psychological research (Boyd et al., 2022). As such, natural language processing, or the sub-field of computer science focused on providing computers with the ability to understand language as humans do (IBM, n.d.), has recently grown in popularity as a way of studying suicide. Naturalistic text data, or language that people produce naturally, is best for this purpose as it is less subject to phenomena like reactivity which can affect the validity of findings.

A number of research groups use natural language processing techniques to study associations between language use and suicidality and to develop methods of detecting suicidality from language use. One natural language processing tool that is useful for psychological research is the Linguistic Inquiry and Word Count, 2022 Version (abbreviated as LIWC-22 or LIWC), a text analysis software that analyzes word use frequency (Boyd et al., 2022). The LIWC takes files of raw text as input and outputs quantitative data about how often they contain words in over 100 dictionaries¹ (also referred to as “categories” or “variables”), which are each made up of groups of words related to a specific aspect of either language style or

¹ A table of all of the dictionary categories included in the LIWC-22, along with descriptions of and sample words contained in them, can be found on pp.11-12 of Boyd et al. (2022), which serves as the LIWC-22 user manual.

content. For example, the pronoun category (titled *pronoun* in the LIWC) contains words that are pronouns such as “I”, “you”, and “it”, while the negative emotion category (titled *emo_neg*) contains words related to the experience of negative emotion such as “bad”, “hate”, and “hurt”. Single words can appear in multiple categories because single words can be related to multiple different language constructs. An example of this is the word “cried”, which is found in 10 different dictionaries including *emotion* (because “cried” suggests sadness), *verb* (because “cried” is a verb), and *focuspast* (because “cried” is in the past tense); this reflects the different ways that the word “cried” can be interpreted to extract meaning from a piece of text.

All LIWC variables, with a few exceptions, are scored as the ratio of words in the variable’s dictionary that are found in the input text to the total number of words in the input text, expressed as a percentage. The *WC* (“word count”) variable is instead scored as a simple count of all of the words in the input text, the *WPS* (“words per sentence”) variable is instead scored as the average words per sentence in the input text, and the summary variables (*Analytic*, *Clout*, *Authentic*, and *Tone*) are instead scored using a proprietary method developed for a previous version of the LIWC by Pennebaker et al. (2015).

This method of quantitatively analyzing text is relatively limited in scope, however, as it fails to take more complex features of language into account. For example, a word (i.e., “sad”) could be preceded by a negation (i.e., “not sad”), which changes the meaning of this expression to its opposite, yet the word would not be removed from the scoring of the category it is included in. Thus, the LIWC is useful for discovering the general qualities of a piece of text but is less useful for gleaning more advanced insights from it. More sophisticated natural language processing techniques have been used to analyze the pre-existing dataset that we use in the present study, as in the CLPsych 2019 Shared Task (Zirikly et al., 2019) that we discuss in more

detail later. As the main goal of our work is to identify differences in linguistic characteristics between suicidal and non-suicidal individuals, we decided that it was not necessary to use more complex procedures.

Suicide Research Using Social Media Data

Social media is an increasingly common source of data for natural language processing research that aims to answer questions about characteristics and detection of mental health issues from text (Shing et al., 2018). One benefit of social media data, as we previously mentioned, is that it is naturalistic; in other words, the statements that people make on social media are natural and are not being interfered with by a lab setting or interaction with researchers. The present study uses data taken from the social media platform Reddit, which is structured as a collection of forums (or *subreddits*, as they are titled on the website) that are each focused around a certain topic and are designed to attract people that are all interested in that topic. A subreddit can also be referred to by its link on the website, in the form of *r/SubredditTitle*. There are subreddits dedicated to a variety of topics, including world news (*r/worldnews*), art (*r/Art*), simple explanations of complicated concepts (*r/explainlikeimfive*), and even psychological research (*r/AcademicPsychology*). Each provides a specialized location for individuals interested in its topic of focus to gather and discuss it. The particular subreddit relevant to the present study is *r/SuicideWatch*, which describes itself as a place of “Peer support for anyone struggling with suicidal thoughts” (*r/SuicideWatch*, n.d.) and is the source of the data we analyze.

Reddit, with the inherent focus on anonymity that is built into the culture of the website, is a great source of data for questions related to sensitive topics like suicide. Anonymity tends to influence people to be more honest than they would be if their identity was known, especially about issues that may be socially undesirable or stigmatizing, like suicidality. One way that

Reddit promotes anonymity is through the idea of a “throwaway” account, or a Reddit account that one creates with the specific intention of posting about something sensitive that they do not want associated with their genuine online identity (Choudhury & De, 2014). Thus, Reddit may encourage honest disclosure more than other social media websites do.

Review of Research on Linguistic Characteristics Associated with Suicidality

Previous empirical research has investigated the characteristics of language that tend to be associated with suicidality using a variety of natural language processing methodologies. Some components of language have been found to be more or less common in text that is produced by suicidal individuals (i.e., have been found to be associated positively or negatively with suicidality); these results and their source studies that we discuss now are summarized in Table 1.

One of the consistent and more intuitive findings in this area of research is that negative emotion words are more frequently found in the language of suicidal individuals than of non-suicidal individuals. Absolutist language, death-related language, self-referential language (such as first-person pronouns), and verbs have also all been found to be used more frequently in the language of suicidal individuals. Additionally, text produced by suicidal individuals tends to contain more words (i.e., have a higher word count) than text produced by non-suicidal individuals. Characteristics of language that have been found to be used less frequently by suicidal individuals than non-suicidal individuals include future-related language and social-reference words. Finally, there are some characteristics of language that research has found mixed results on; some studies have found that suicidal individuals tend to use both adverbs and prepositions more often than non-suicidal individuals, while others have found that these language categories are used less by suicidal individuals than non-suicidal individuals.

Table 1*Summary of Reviewed Research on Linguistic Characteristics Associated With Suicidality*

Language Characteristic	Association With Suicidality	Supporting References
absolutist language	more common	Al-Mosaiwi & Johnstone (2018), Homan et al. (2022)
death-related language	more common	Homan et al. (2022), Liu et al. (2019), Stirman & Pennebaker (2001), Zhang et al. (2021)
negative emotion words	more common	Al-Mosaiwi & Johnstone (2018), Barak & Miron (2005), Brown et al. (2019), Homan et al. (2022), Howard et al. (2020), Kim et al. (2019), Lekkas et al. (2021), Stirman & Pennebaker (2001), Zhang et al. (2021)
self-referential language	more common	Barak & Miron (2005), Brancu et al. (2016), Homan et al. (2022), Kim et al. (2019), Stirman & Pennebaker (2001), Zhang et al. (2021)
verbs	more common	Homan et al. (2022), Litvinova et al. (2017), Pestian et al. (2008)
word count	higher	Brancu et al. (2016), Homan et al. (2022), Pestian et al. (2008)
future-related language	less common	Homan et al. (2022), Kim et al. (2019), Liu et al. (2019)
social-reference words	less common	Gunn & Lester (2012)

adverbs	mixed	Homan et al. (2022), Kim et al. (2019), Zhang et al. (2021)
prepositions	mixed	Homan et al. (2022), Litvinova et al. (2017), Pestian et al. (2008)

The source of the text analyzed in most of these studies was either social media or online forums/blogs, similar to the present study (Al-Mosaiwi & Johnstone, 2018; Brown et al., 2019; Gunn & Lester, 2012; Howard et al., 2020; Lekkas et al., 2021; Litvinova et al., 2017; Liu et al., 2019), although others used text from poetry (Stirman & Pennebaker, 2001), online support groups (Barak & Miron, 2005), personal documents such as suicide notes and diaries (Kim et al., 2019; Pestian et al., 2008; Zhang et al., 2021), or a qualitative suicide risk assessment measure (Brancu et al., 2016). In addition to these individual studies, the meta-analysis conducted by Homan et al. (2022) analyzed text from a variety of sources, including clinical populations, student populations, veteran populations, and social media and online forums; some of the separate studies we discuss were also included in this meta-analysis.

The case study conducted by Gunn & Lester (2012) illustrates the variability of language characteristics used by suicidal individuals and demonstrates how these are only general patterns. They analyzed the Twitter posts made by a woman in the 24-hour period leading up to her death by suicide and found that she used more social-reference words, fewer self-reference words, and fewer negative emotion words as the time of her death approached. Some of these findings are inconsistent with the general consensus of the research we discuss above; for example, this woman used fewer negative emotion words as her death approached, but this is unexpected based on the findings of most other research into this topic. However, this is potentially because

this study looked into only one suicidal individual's language, and her pattern of language use happens to differ from that of the average suicidal individual. It is also likely that this study investigated a phenomenon that is fundamentally different than that studied in other work, where it is unknown if the participants later died by suicide; Gunn & Lester (2012) looked at a case of suicide as it unfolded, and the imminence of death may mean that the language used by this woman is different than that used by people who are suicidal yet are not actively planning on committing suicide in the near future.

We expect that the ideas reflected in the theoretical and empirical work that we review previously will show up in the language of suicidal individuals more often than in the language of non-suicidal individuals. As suggested by the research we summarize in Table 1, LIWC scores on variables related to absolutist language, death-related language, negative emotion words, self-referential language, verbs, and word count are likely to be higher in text produced by suicidal compared to non-suicidal individuals, while LIWC scores on variables related to future-related language and social-reference words are likely to score lower. The work of both Baumeister (1990) and Shneidman (1993, 1998) further support the ideas that all-or-nothing language and words related to negative emotion may be more common in text produced by suicidal individuals compared to non-suicidal individuals, as these language dimensions likely reflect the states of cognitive deconstruction and mental pain, respectively. Finally, fatigue-related words may be used more frequently by suicidal individuals compared to non-suicidal individuals, potentially because of a feeling of being tired of suffering and a desire to escape as posited by Baumeister (1990). These ideas guided our process of selection of the 10 LIWC variables that we expected to differentiate well between suicidal and non-suicidal individuals in our data, which are *allnone* (all-or-none thinking), *auxverb* (auxiliary or helping verbs), *death* (death), *emo_neg* (negative

emotion), *fatigue* (state of fatigue), *focusfuture* (future-focused words), *i* (first-person singular pronouns), *verb* (common verbs), *we* (first-person plural pronouns), and *WC* (total word count).

Ethical and Methodological Concerns Related to Suicide Research and Social Media Data

Although studying suicide is a demanding endeavor with many ethical, methodological, and moral complications, it is essential if we are to gain a deeper understanding of the phenomenon. Just as it is important to raise awareness about suicide on the individual and societal levels so that we may bring attention to its presence and impact on people, it is also important to conduct research on suicide so that we may learn more about it and develop ways to help people who are struggling with suicidality. Excluding uncomfortable topics from study is not how we should proceed with science, as it prevents us from investigating what are often the most critical and profound questions with the most impact on human life. Put differently, science does not move forward by ignoring difficult problems; it does so by engaging with them.

However, as suicide is such a controversial and delicate topic, the need for knowledge about suicide and the need to study it in a considerate and sensitive manner must be balanced with each other. Care needs to be taken when speaking about and investigating suicide in order to protect the wellbeing of research participants, researchers, and the general public as much as possible (Chancellor et al., 2019). Additionally, it is critical to pay attention to the benefits and disadvantages of suicide research so that we design studies that minimize risk to participants and those to whom the results of the research could be applied. As the present study uses a pre-existing archive of Reddit posts as its data source, we discuss these considerations about suicide research as they relate to and interact with the use of social media data, which comes with its own set of ethical and methodological concerns.

The disadvantages of suicide research with social media data are complicated to both identify and address. Ethical drawbacks include privacy, informed consent, the potential for harmful results and applications, dealing with the stigma surrounding suicide, and the idea of “suicide contagion,” while methodological challenges include the question of whether work that uses social media data should be considered human subjects research (as opposed to public records research) and issues of reliability and validity.

The experience of suicidality is often upsetting and private, and can have consequences on many levels including personal, social, and professional. As such, information about suicidal individuals should be treated with great care to ensure that any research conducted with suicidal individuals prioritizes their privacy and mental well-being. Precautions to ensure privacy might include de-identifying any materials linked to or produced by participants and not sharing these materials with anyone outside of the team directly involved in the research (Benton et al., 2017). The use of public social media data complicates this further, as social media posts are realistically accessible to anyone with an Internet connection yet may contain information that could be used to identify the person who posted them. Furthermore, researchers and social media users may perceive the visibility and privacy of social media data differently: researchers may believe that social media posts are public information because they are publicly accessible, yet social media users may believe that their posts are private because they do not expect a large audience to view their posts. Thus, social media data should always be treated as though it contains private information; this requirement becomes even more essential when the data is already known to contain sensitive information, as some of the data we use in the present study does because it comes from the suicide support forum on Reddit.

Informed consent is also a critical issue in the context of suicide research using social media data, as it is ideal that people know about and agree to the fact that their online behavior on such a sensitive topic is being used as data in research. However, in research that takes advantage of the large amount of data available online, as the present study does, obtaining informed consent is not always possible due to the practical difficulty of individually receiving consent from each participant in a sample of potentially thousands or millions of social media users (Benton et al., 2017). Informed consent should be obtained whenever possible to ensure that we use data in line with the wishes of those who produce it, but this is often a difficult task when using large-scale samples of social media data.

Additionally, it is possible that the results of suicide research could be used by individuals who seek to cause harm, regardless of how well-intentioned researchers are (Chancellor et al., 2019). For example, an individual could use information about characteristics of suicidality to seek out and harass suicidal people online, or an employer could discriminatively screen social media profiles of potential employees for characteristics of suicidality and refuse to hire those who display them as they might be construed as impediments to productive work. However, most knowledge can be used to either do harm or do good; thus, we must carefully consider the potential negative uses of knowledge we might gain from a specific line of research, weigh these consequences against the consequences of lacking this knowledge, and make an appropriate decision.

Concerns related to social processes, including social stigma and social contagion, are also relevant to the study of suicide. Suicide is an extremely sensitive and distressing topic that carries social stigma, which involves fear, misunderstanding, and ignorance of suicide (Yeskuatov et al., 2022). Suicidal people are often acutely aware of this stigma, which prevents

many from reaching out for help from people they are close to or from mental health professionals, which in turn leads many to attempt to manage their suicidal thoughts and behaviors alone. By conducting research on suicide, we bring attention to and encourage discourse about this stigmatized topic, but we may also desensitize ourselves to a painful topic on a societal level. The way in which we discuss suicide also matters. Certain methods of talking about suicide (particularly those that simplify, sensationalize, or inadvertently glorify the phenomenon) can be dangerous as they may encourage “suicide contagion”, or the process through which exposure to suicide may lead to suicidal behavior in those exposed (O’Carroll & Potter, 1994). It is true that discussion of and research on suicide can cause unintended discomfort or harm. However, learning more about suicide and reaching out to people who may need help in a way that acknowledges yet does not reinforce what they are struggling with is worthwhile and constructive.

One major methodological question in the area of social media research is whether or not the use of public social media data should be classified as human subjects research. From one perspective, social media data is accessible by the public, but from the other, social media data is often believed to be somewhat private by social media users and may contain information they do not want disclosed in a research context. Some users even fundamentally misunderstand basic characteristics of the Internet, such as the fact that posts made to the Internet are essentially permanent (Mikal et al., 2016, as cited in Benton et al., 2017). Many ethics boards have responded to this debate by considering public social media data to be analogous to public records data and requiring researchers to treat it as such in research approval and institutional review settings (Chancellor et al., 2019). However, if this consensus changes and we come to

define social media research as human subjects research, this will alter ethics protocols such as those surrounding informed consent, which will become necessary to receive from participants.

Additionally, we need to be sure whether the screening and diagnostic tools that could be developed through social media research into the prediction of suicide are measuring what we think they are, especially before they are applied to practical use. It is possible that, for example, a screening tool using an algorithm developed with social media data could be good at detecting suicidality only as it manifests in the population of typical social media users (i.e., people who are younger and more adept with technology; Chancellor et al., 2019). A tool developed using data from a support forum, as the present study uses, might overwhelmingly rely on data from the type of person who is willing to acknowledge and talk about their psychological difficulties. There is also the potential for an algorithm to misclassify people, detecting either false positive cases (i.e., detecting suicidality when it is actually absent) or false negative cases (i.e., failing to detect suicidality when it is actually present). The cost of these mistakes (particularly false negatives) that could be made by computerized screening tools are very high, so we must be cautious when using them and only apply them to real situations once we know that they are capable of classifying well enough to justify their use.

Despite these potential drawbacks, the benefits of research into suicide are apparent: we gain improved knowledge about suicide, become more able to detect the presence of suicidality, and develop new prevention and treatment methods, all of which are likely to reduce the suffering and public health impacts associated with suicide. There are also some specific advantages tied to the approach that we take in the present study. We analyze naturalistic text data produced by Reddit users, which is a methodological benefit because the production of this data is unprompted and thus is more likely to reflect genuine characteristics of the users; this is

important when researching a phenomenon that is stigmatized, such as suicide. Additionally, as we use archival text data that had already been produced by the time it was sampled, we avoid the risk of distressing participants or exposing them to the idea of suicide ourselves, as more contemporaneous and non-archival methods might do.

The Present Study

We analyze a pre-existing dataset of Reddit posts and metadata, the University of Maryland Reddit Suicidality Dataset, Version 2 (Shing et al., 2018; Zirikly et al., 2019), to investigate the linguistic characteristics associated with suicidality. The aims of this study are twofold. First, we intend to replicate previous research findings about the linguistic characteristics associated with suicidality and test new hypotheses about these characteristics based on psychological theory and research. Second, we aim to further the integration of the psychological literature on suicide with work from computer scientists on linguistic characteristics associated with suicidality. Our dataset has mostly been used for the latter type of research, as in the CLPsych 2019 Shared Task (Zirikly et al., 2019), which was a set of suicide detection and prediction tasks that teams of researchers participated in to develop classifiers for the groups (i.e., suicidal or non-suicidal) within the dataset. While the CLPsych 2019 Shared Task appeared to be focused around differentiating between suicidal and non-suicidal individuals for the purpose of being able to differentiate, we are more interested in what the qualities by which they are differentiated can tell us about suicidality. Thus, we wanted to conduct our work in a way that ensured we took the psychological literature on suicide into account.

We hypothesize that the language used by suicidal individuals on Reddit (operationalized as users who posted to r/SuicideWatch) will differ from the language used by non-suicidal individuals on Reddit (operationalized as users who posted to Reddit but never to

r/SuicideWatch or any of a certain set of other subreddits related to mental health). We also developed a list of LIWC-22 variables, based on the theory and research that we reviewed, that we believe will reflect the language characteristics used more or less often by suicidal compared to non-suicidal Reddit users in our dataset. Thus, we also believe that these variables will differentiate between suicidal and non-suicidal users in our dataset if used as predictors in a regression model; these variables were *allnone*, *auxverb*, *death*, *emo_neg*, *fatigue*, *focusfuture*, *i*, *verb*, *we*, and *WC*.

Method

Data

The source of our data for the present study was the University of Maryland Reddit Suicidality Dataset, Version 2 (Shing et al., 2018, Zirikly et al, 2019), which was generously shared by the University of Maryland research team responsible for creating it and the American Association of Suicidology.² Hereafter we refer to this dataset as the “UMD dataset.” IRB approval was obtained for this study. Post title and post body text, along with metadata such as post ID numbers and user ID numbers, from a total of 2,053,200 Reddit posts from 21,767 Reddit users are included in the UMD dataset. The Reddit users in the dataset can be divided into two groups: a suicidal group, composed of all Reddit users who made at least one post to r/SuicideWatch, and a control group, composed of all Reddit users who made no posts to r/SuicideWatch or to any of a set of specified subreddits related to mental health originally selected by Pavalanathan and De Choudhury (2015).³

² Additional information about the dataset can be found at http://users.umiacs.umd.edu/~resnik/umd_reddit_suicidality_dataset.html.

³ The full list of subreddits excluded from the control group, as listed by Shing et al. (2018) and originally developed by Pavalanathan and De Choudhury (2015) is as follows: r/addiction, r/alcoholism, r/Anger, r/bipolarreddit, r/BPD (borderline personality disorder), r/depression, r/DPDR (depersonalization-derealization disorder), r/EatingDisorders, r/feelgood, r/getting_over_it, r/hardshipmates, r/mentalhealth, r/MMFB (Make Me Feel

The UMD dataset is sampled from the full contents of Reddit between January 1, 2008 through August 31, 2015 and the partial contents of Reddit from 2006 and 2007. The sample is made up of all Reddit posts by any Reddit user who had posted to r/SuicideWatch and had made at least 10 posts to Reddit, along with all Reddit posts from an equal number of users who had not posted to r/SuicideWatch or any of the set of specified subreddits related to mental health. Thus, both the suicidal group and control group contained posts from across Reddit, instead of the suicidal group containing only posts from r/SuicideWatch and the control group containing posts from all other subreddits in the data. All original posts made by each user were included in the data, but comments on other users' posts were not.

A Reddit user in the UMD dataset is classified into the suicidal group if they had a Reddit post history that contained any posts in r/SuicideWatch. Shing et al. (2018) explain that the rationale for operationalizing suicidality in this manner is based on the approach taken for a similar task by Coppersmith et al. (2014), who identified Twitter users with mental disorders by searching for those who had made posts containing a self-reported diagnosis of a mental disorder (e.g., "I was diagnosed with X "). A self-reported diagnosis was treated as a signal that the user actually had the disorder; in the same way, self-reporting suicidality by choosing to post to r/SuicideWatch was treated as a signal that a user in our dataset was actually suicidal. However, this assumption is unlikely to be correct in all instances, such as in cases where an individual posts to r/SuicideWatch seeking help for a friend or relative who is suicidal; one such example of this from our data is a post to r/SuicideWatch titled "Desperate need of help, friend of a friend is in trouble".

Better), r/panicparty, r/psychoticreddit, r/ptsd, r/rapecounseling, r/schizophrenia, r/socialanxiety, r/StopSelfHarm, r/SuicideWatch, r/survivorsofabuse, r/traumatoolbox.

Analyses

After receiving the UMD dataset and conducting a cursory visual inspection of it to get a general idea of its structure (i.e., to learn what specific information was contained in the dataset and how it was organized), we set aside our data until we finished writing our preregistration, or plan for how exactly we were going to analyze our data. This preregistration, along with other materials related to the present study such as the code documents we used for our analyses, can be found on in our Open Science Framework (OSF) repository at https://osf.io/6872d/?view_only=ae805a3c25f94be6ad72720234542cf5. This thesis includes a subset of the analyses proposed in the preregistration due to time constraints. All analyses were conducted using the R programming language (v. 4.2.2; R Core Team, 2022) and the Linguistic Inquiry and Word Count, 2022 Version (Boyd et al., 2022). Our plan of analysis was composed of three major steps: preprocessing, processing of the raw text data using the LIWC-22, and analysis of the LIWC-22 output data.

Preprocessing

The raw post text data was originally in two separate data files, which was due to the training-test division of the data made by the research team for the CLPsych2019 Shared Task, so we first combined both of these data files into one larger dataset that contained all post text and metadata. We also combined the post title and post body text into one observation per user in order to work around the issue of some posts containing title text but no body text. We then removed any observations within the dataset that contained NA (missing) values and any instances of de-identifiers from within the text. In our data, de-identifiers are text tags that the original researchers used to replace any identifying terms included in the raw text data, such as “_PERSON_” and “_URL_” which were used to replace the names of people or URLs in the

text, respectively. After preprocessing of our raw data, we were left with a total of 2,045,362 Reddit posts from 21,741 Reddit users.

Analysis of Raw Text Data and LIWC Output Data

We next processed our reduced raw text data using the Linguistic Inquiry and Word Count, 2022 Version, to obtain LIWC scores as output. Afterwards, we aggregated the LIWC scores by user so that we could conduct our analyses at the user level instead of the post level, which is how our data was originally structured. To aggregate, we converted the LIWC scores that are simple percentages to raw counts, summed these counts for each user, and then divided these counts by the value of *WC* (word count) for each user to obtain the average weighted by word count for each score for each user. The *WPS* (average words per sentence) variable thus became *avgWPS* (the “average of the average” of words per sentence) for each user, and as it is unclear how the summary variables *Analytic*, *Clout*, *Authentic*, and *Tone* are scored we treated them as percentages as well. We also reverse-coded the *focusfuture* (future-focused language) variable in this step, as the evidence we reviewed suggested that this variable was the only one of our 10 selected variables that would score lower in the suicidal group compared to the control group. Finally, we removed from our data any users that posted fewer than 100 words, as we thought that the low word count might introduce confounding variability in our results. After aggregation, each observation in our dataset corresponded to the set of aggregated LIWC scores for one unique Reddit user’s posts.

Next, we standardized a selected set of the LIWC variables to a mean of 50 and a standard deviation of 10 to allow for efficient and meaningful comparison of the effect sizes of the variables that were measured differently (i.e., raw counts vs. percentages vs. some unknown proprietary calculation method). These variables included: *AllPunc* (all punctuation), *Analytic*

(logical/formal thinking), *Apostro* (apostrophes), *Authentic* (honesty/genuineness), *avgWPS* (average of the average words per sentence), *Clout* (leadership/status), *Comma* (commas), *Exclam* (exclamation points), *OtherP* (other punctuation marks), *Period* (periods), *QMark* (question marks), *Tone* (degree of positive/negative tone), and *WC* (total word count). We then split the aggregated and standardized LIWC output data into training and test samples for use in our logistic regression analyses; 60% of our data became model training data and 40% of our data became model testing data, using a seed for the sake of reproducibility. Finally, we created a dummy-coded variable called *fromSW* (“from the r/SuicideWatch group) to tag each user for whether they were part of the control group (*fromSW* = 0) or the suicidal group (*fromSW* = 1).

Our first step in analyzing the LIWC-22 output data was to calculate the zero-order correlations between *fromSW* and all 118 aggregated and standardized LIWC variables. Next, we built two logistic regression models to test our hypotheses, using the aggregated and standardized LIWC scores as our predictor variables and *fromSW* (i.e., whether the user was in the suicidal or control group) as our outcome variable. Our first model, Model 1, used all 118 LIWC variables as predictors, while our second model, Model 2, used only our 10 selected LIWC variables (*allnone*, *auxverb*, *death*, *emo_neg*, *fatigue*, *focusfuture*, *i*, *verb*, *we*, and *WC*) as predictors. We used the *tidymodels* package for R (v. 1.0.0; Kuhn et al., 2020) to build our models. Finally, we investigated the best potential predictor variables for building a logistic regression model (i.e., we conducted a model selection analysis) using the *regsubsets()* function from the *leaps* package for R (v. 3.1; Lumley, 2020). This function conducted an exhaustive stepwise analysis that extracted the set of best predictor LIWC variables of the outcome of interest, *fromSW*.

Results

The mean of Reddit posts per user was $M = 96.46$, and the standard deviation of Reddit posts per user was $SD = 628.15$. The mean of word count per post was $M = 47.57$, and the standard deviation of word count per post was $SD = 113.48$. The mean of word count per user was $M = 4604.26$, and the standard deviation of word count per user was $SD = 19022.14$. Investigating the class imbalance of our data revealed that there were about the same number of control-group users ($n = 10,250$) as suicidal-group users ($n = 10,872$), but that there were about three times as many posts made by the suicidal group ($n = 1,513,032$) as posts made by the control group ($n = 532,203$).

The 10 strongest correlations of both directions between the dichotomous *fromSW* variable and all LIWC variables are displayed in Table 2, along with sample words contained in the LIWC dictionary categories captured by the variables.

Table 2

Strongest Correlations Between fromSW and All LIWC Variables

LIWC Variable	Description	Sample Words	Correlation With <i>fromSW</i>
<i>Analytic</i>	Analytical thinking	N/A	-0.506
<i>ppron</i>	Personal pronouns	I, you, my, me	0.503
<i>pronoun</i>	Total pronouns	I, you, that, it	0.499
<i>i</i>	First person singular pronouns	I, me, my, myself	0.491
<i>negate</i>	Negations	not, no, never, nothing	0.461
<i>Linguistic</i>	Linguistic dimensions	N/A	0.456

<i>function</i>	Total function words	the, to, and, I	0.455
<i>Dic</i>	Words in LIWC dictionaries	N/A	0.446
<i>verb</i>	Common verbs	is, was, be, have	0.445
<i>emo_neg</i>	Negative emotion	bad, hate, hurt, tired	0.424

Note. LIWC variables marked with “N/A” are described without sample words: *Analytic* = summary variable computed using a proprietary formula developed by Pennebaker et al. (2015) that measures logical/formal thinking; *Linguistic* = variable comprised of all linguistic variables measured by the LIWC; *Dic* = percentage of all words that are found in dictionaries in the LIWC.

Of the 118 variables used as predictors of *fromSW* in our first logistic regression model (Model 1), 35 were significant at the $p < .05$ level and 21 were significant at the $p < .01$ level. Of the 10 selected LIWC variables used as predictors of *fromSW* in our second logistic regression model (Model 2), 8 were significant at the $p < .05$ level and 7 were significant at the $p < .01$ level. The best predictors of *fromSW* in both directions are displayed in Table 3 (for Model 1) and Table 4 (for Model 2), along with regression coefficients for each variable and sample words contained in the LIWC dictionary categories captured by the variables.

Table 3

Logistic Regression Model 1: Best Predictors of fromSW Significant at $p < .05$

LIWC Variable	Description	Sample Words	Coefficient
<i>mental</i>	Mental health	mental health, depressed, suicid*, trauma*	4.143
<i>death</i>	Death	death*, dead, die, kill	1.202
<i>filler</i>	Filler words	rr*, wow, sooo*, youknow	1.064
<i>swear</i>	Swear words	shit, fuckin*, fuck, damn	0.816

<i>socrefs</i>	Social referents	you, we, he, she	-0.728
<i>Social</i>	Social processes	you, we, he, she	0.715
<i>socbehav</i>	Social behavior	said, love, say, care	-0.700
<i>tone_neg</i>	Negative emotional tone	bad, wrong, too much, hate	0.559
<i>we</i>	First person plural pronouns	we, our, us, lets	-0.496
<i>conflict</i>	Interpersonal conflict	fight, kill, killed, attack	-0.480

Note. Sample words that end in an asterisk are word stems that lead to any word with that stem being counted for the LIWC category they are in (i.e., “suicid*” being in the *mental* LIWC category results in both “suicide” and “suicidal” being counted towards this category).

Table 4

Logistic Regression Model 2: Best Predictors of fromSW Significant at $p < .05$

LIWC Variable	Description	Sample Words	Coefficient
<i>death</i>	Death	death*, dead, die, kill	1.531
<i>emo_neg</i>	Negative emotion	bad, hate, hurt, tired	1.193
<i>allnone</i>	All-or-none thinking	all, no, never, always	0.387
<i>WC</i>	Word count	N/A	0.321
<i>i</i>	First person singular pronouns	I, me, my, myself	0.292
<i>auxverb</i>	Auxiliary (helping) verbs	is, was, be, have	0.139
<i>we</i>	First person plural pronouns	we, our, us, lets	0.123
<i>verb</i>	Common verbs	is, was, be, have	0.039

Note. Sample words that end in an asterisk are word stems that lead to any word with that stem being counted for the LIWC category they are in (i.e., “death*” being in the *death* LIWC category results in both “death” and “deathly” being counted towards this category).

The confusion matrices for each of our two models, which show how many instances of true positive, true negative, false positive, and false negative predictions our models made for which group a Reddit user was in, are displayed in Table 5.

Table 5*Confusion Matrices for Models 1 and 2*

Model 1 Confusion Matrix		
	Actual Control Group	Actual Suicidal Group
Predicted Control Group	true negative ($n = 3457$)	false negative ($n = 769$)
Predicted Suicidal Group	false positive ($n = 642$)	true positive ($n = 3581$)
Model 2 Confusion Matrix		
	Actual Control Group	Actual Suicidal Group
Predicted Control Group	true negative ($n = 3294$)	false negative ($n = 840$)
Predicted Suicidal Group	false positive ($n = 805$)	true positive ($n = 3510$)

We assessed the predictive abilities of our two models using the model evaluation statistics of accuracy, sensitivity, specificity, and the kappa value. The values of these four statistics for each of our two models are presented in Table 6.

Table 6*Model Evaluation Statistics for Models 1 and 2*

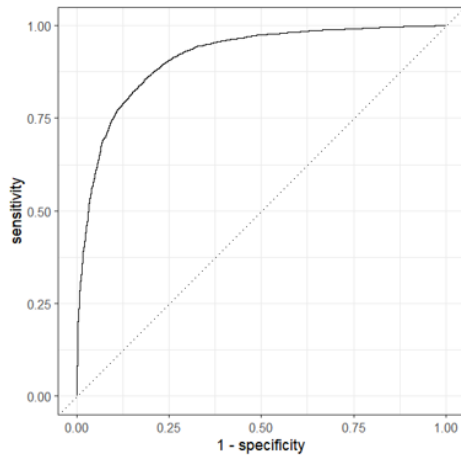
Statistic	Model 1	Model 2
accuracy	0.83	0.81
sensitivity	0.84	0.8
specificity	0.82	0.81
kappa value	0.67	0.61

Note. These model evaluation statistics are defined as follows: *accuracy* = proportion of data points predicted correctly; *sensitivity* = ratio of cases correctly classified as positive to actual positive cases; *specificity* = ratio of cases correctly classified as negative to actual negative cases; *kappa value* = indicator of how much better the model classifies compared to a model that classified randomly based on the frequency of each class's occurrence.

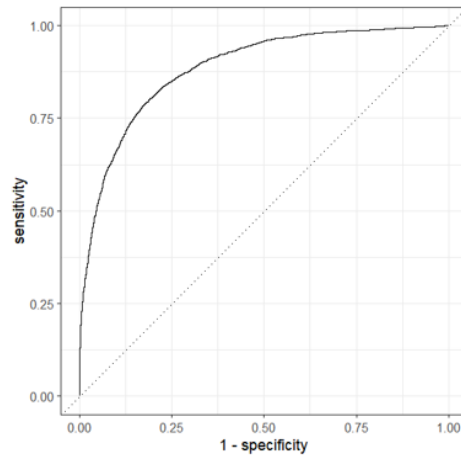
For each of our two models, we graphed the receiver operator curves (ROC) and calculated the area under the receiver operator curves (AUC), which are displayed in Figure 1 and Table 7, respectively.

Figure 1

Receiver Operator Curves (ROC) for Models 1 and 2



ROC for Model 1



ROC for Model 2

Table 7

Area Under the Receiver Operator Curve (AUC) Statistic for Models 1 and 2

	Model 1	Model 2
Area under the receiver operator curve (AUC)	0.91	0.89

The best potential predictor variables out of all 118 LIWC variables returned for a logistic regression model through our model selection analysis were *Analytic* (marked on step 1), *pronoun* (marked on step 2), *mental* (marked on steps 2 and 3), *ppron* (marked on step 3), and *negate* (marked on step 3).

Discussion

Discussion of Results

Taken together, the results of the present study indicate that the language used by suicidal and non-suicidal individuals on Reddit differs, which supports our main hypothesis and is consistent with the findings of other studies on this topic. Additionally, our work directly replicated some of these specific findings and found support for the theoretical ideas that we reviewed, both of which we will discuss in more detail later. The models that we developed also performed well at distinguishing between the suicidal and control groups in our data, demonstrating that computerized detection of suicidality is possible and can be performed with a relatively high level of accuracy. We also noticed that the results of our zero-order correlations were somewhat different from the results of our logistic regression models. This reflects the fact that the LIWC variables are often highly intercorrelated (i.e., they measure similar constructs or they share words) and so they share variability, which could have affected these results.

About one-third of the 118 LIWC variables that we used as predictors in our first logistic regression model and about three-quarters of the 10 LIWC variables that we used as predictors in our second logistic regression model were able to significantly predict whether Reddit users were in the suicidal or control group. As this is a higher number of variables than would be expected to be statistically significant by chance, we can be relatively confident that our results are demonstrating that there are significant differences between the language of suicidal and non-suicidal individuals. This indicates that the characteristics of the language that a suicidal person produces on Reddit tend to differ quantifiably from the characteristics of the language of non-suicidal individuals on Reddit. Moreover, this suggests that the language dimensions captured by the Linguistic Inquiry and Word Count are capable of differentiating between individuals who

are and are not suicidal. The specific variables that best differentiated between our two groups included variables related to both language content (such as *mental*, *death*, and *Social*) and language style (such as *filler*, *swear*, and *we*), suggesting that the main differences between the language used by suicidal and non-suicidal individuals are in both content and style. In other words, suicidal individuals appeared to both write about different things (i.e., language content) and write in different ways (i.e., language style) compared to non-suicidal individuals on social media.

To further illustrate the differences in language use between the suicidal and control groups in our data that are reflected by differences in LIWC variable scores, we manually investigated the post profiles of the Reddit users in the suicidal group that scored lowest and highest on some of these variables. We chose to limit this to only users in the suicidal group because, when we looked at users from both groups at first, we noticed that the extreme scores usually were from bots and thus did not provide us with meaningful information about how people on Reddit use language. We assumed that users that had posted to r/SuicideWatch were more likely to be real people for two reasons: the subreddit has relatively low user traffic (417,000 members as of April 30, 2023), which seems to indicate that bots are less common on it than larger subreddits, and it seems less likely that bots would choose to post to a suicide support subreddit.

We found that the variables most strongly related to the suicidal group tended to be picking up on aspects of language such as expression of a wish to die, indications of self-harm, or discussion of interpersonal difficulties. Phrases that were used by some of these users included “I want a hug but I have no one to give one”, “I feel like doing nothing”, “You have no idea what it’s like”, “All I want to do is die”, and “Always alone”. The suicidal group in our data posted

about their experiences with suicidality in r/SuicideWatch, which often involves discussion of mental health issues or death; as such, they were using more *mental* and *death* words. Negative emotion and tone (*emo_neg* and *tone_neg*, respectively) reflect the suicidal group's tendency to discuss the experience of negative emotions, which is likely related to their experience of mental pain. Words related to social processes, as in the variables *socrefs*, *Social*, and *socbehav*, are likely being used in the context of interpersonal struggles, such as interpersonal conflicts that may be motivating factors behind mental pain. For example, a person may have experienced a social interaction that triggered feelings of suicidality in them and thus they wrote about that on Reddit. They might, however, be writing about how the people in their life are reacting to their feelings of suicidality. All-or-none language may be more common on the part of suicidal individuals because the state of cognitive deconstruction that may be characteristic of suicidality is reflected by a reduced ability to perceive nuance in decision-making.

The variables that we just discussed are more related to linguistic content, yet others that differentiated between the two groups were related to linguistic style. The finding that the suicidal group posted more words (i.e., had a higher word count) may be a result of the fact that suicidal individuals tend to try to express their mental state and what they are experiencing to others. They may also be turning to Reddit as an outlet because they feel that they have nowhere else to turn to discuss the feeling of suicidality. The suicidal group in our dataset may have been either writing longer posts or posting more frequently, both of which would indicate an increased volume of writing produced and thus a greater attempt to express themselves than the control group. However, the higher word count on the part of the suicidal group could have also been an artifact of the nature of people who choose to post about feeling suicidal online; they may simply be more willing to share things online, especially sensitive things, or they may spend more time

online because they are choosing to turn to online support groups instead of people in their offline lives. Other style variables, such as negations and self-referential language, may be used because of the nature of discussing suicidality; for example, negations may be more common because phrases like “not worth it” are used more frequently, and self-referential language may be used more often because speaking about one’s own experience of suicidality inherently involves referencing the self.

Some of these specific findings (i.e., suicidal individuals tend to use more death words, more negative emotion words, more language related to all-or-nothing thinking, more self-referential language, and more words in general) are consistent with previous empirical research into this topic. Some of the past empirical research that we reviewed previously that investigated linguistic characteristics of suicidality found these results as well, and as such we have replicated those findings. Additionally, our results appear to be consistent with the theoretical orientations to suicide that we reviewed (Baumeister, 1990; Shneidman, 1993, 1998), which argue that suicidal individuals are motivated to commit suicide because of extreme mental pain and that suicidality leads a person into a state of narrowing of consciousness or cognitive deconstruction. Suicidal individuals in our data tend to discuss the mental pain that they are experiencing, and they also tend to talk about it often in order to try to work through and express their experiences.

Our models were overall quite effective in classifying users as either suicidal or non-suicidal, as both of our models predicted successfully at a level higher than chance (~ 80% for each of our models). They were able to classify a user into the correct group (i.e., the group that they actually were a member of) a majority of the time, which has implications for computerized detection of suicidality and other mental health issues. Additionally, the receiver operator curves and AUC (area under the receiver operator curve) statistics indicated that our models performed

well at classifying across a range of classification thresholds. In other words, if we wanted to make our model incorrectly classify in the direction of false positives more often (i.e., decide that a user is suicidal when they are actually not) by lowering the decision threshold for this because we decide that making the mistake of saying a person is suicidal when they are actually not will lead to the detection of more actually suicidal cases, we can be relatively confident that it will perform well here because of what these two measures tell us.

Developing an automatic classifier using techniques similar to those we used could be simultaneously beneficial and harmful. From one perspective, developing a highly accurate method of classifying individuals for suicide risk could be extremely useful for practical application in settings where there is a high volume of patients and the relative cost of not treating patients who need urgent help is high (e.g., in an emergency room). A classifier could also be used as a supplementary quantitative measure in the process of psychotherapy intake or as a participant differentiation method for research studies. In these scenarios, a person could be asked to produce a writing sample to be scored for suicide risk, the results of which could help direct them to the appropriate clinician for their needs or place them into the correct experimental group. It is also possible, though, that classifiers of this sort could be used to harass or otherwise harm the individuals that they would presumably detect. If a bad actor were able to screen social media data for individuals at risk for suicide quickly and with relatively few resources, for example, they could target and attempt to harm these people, who may be at risk for impulsive self-threatening behavior. The ability to infer the mental state of a person from the language that they use is a delicate one and should be used only by trustworthy individuals and groups that plan to help those that they identify using these methods.

There is also the general risk that classifiers of any sort may make errors, in either the false positive (detecting the presence of some trait when it is actually not present) or false negative (failing to detect the presence of some trait when it is actually present) direction. Each of our models made errors in classifying users in our dataset around 20% of the time, which has implications for the responsibilities that we should give to such classifiers. In the context of detecting suicidality, a false negative error (which Model 1 made around 9% of the time and Model 2 made around 10% of the time) seems to be more of a consequential and potentially fatal mistake than a false positive error (which Model 1 made around 8% of the time and Model 2 made around 10% of the time), but we should attempt to reduce the occurrence of both types of errors as much as possible. If a classifier makes a false-negative error about a person, this is a major oversight as the genuine presence of suicidality has gone undetected and a potentially preventable death may occur. If a classifier makes a false-positive error about a person, they may experience unnecessary distress prompted by being told that they match the general profile of a suicidal person or could have existing (yet minor) psychopathological symptoms exacerbated simply due to exposure to this idea. Thus, it is critical that classifiers are as accurate as possible if we choose to use them for practical applications.

Prediction of suicidality using language characteristics can also inform us about the nature of suicide. By attempting to predict whether users in our dataset were suicidal or non-suicidal, we investigated how a computer model was making these decisions and thus obtained information about what suicidal individuals are like compared to non-suicidal individuals. We found information about the composition of the language that suicidal individuals tend to use, which in turn informs us of the psychological traits of suicidal individuals on average and about how they tend to express themselves. Thus, the results of research into prediction of suicidality

using linguistic characteristics go two ways: they can provide us with information on how to predict suicidality because we develop models that can complete this task, and they can tell us about the characteristics of suicidal individuals by nature of what these predictive characteristics are.

Limitations and Future Directions

Previous research into linguistic characteristics associated with suicidality has mostly remained in the domain of computer science and has mainly focused on how to classify text for the presence of suicidality or for suicide risk. As such, there has been comparatively little focus on psychology in this area of research. The present study contributes uniquely to this area by integrating our empirical findings with psychological explanations. However, our work is limited in a few major ways, which we now discuss: (1) the outcome of interest, suicidality, was measured in a manner with relatively low validity; (2) some components of the theories we used to guide our work do not have direct variable correlates in the LIWC; and (3) the LIWC offers relatively limited insight into language use compared to other, more sophisticated natural language processing techniques.

First, our outcome of interest, suicidality, was measured in a manner with relatively low validity by whether or not Reddit users had posted to r/SuicideWatch, the suicide support forum on Reddit. The source of our data was thus a non-clinical sample, and we also had no information on whether any of the Reddit users in our dataset later died by suicide. There is also “noise” in the data, such as instances of posts made by bots or of users posting to r/SuicideWatch when they are not actually expressing suicidality. Future research may ameliorate these issues by analyzing text data from individuals that have emitted a more valid signal of suicidality, such as by having entered psychotherapy and being assessed as suicidal by a clinician. The dataset that

we used also contains expert and crowdsourced suicide risk ratings for a subset of its Reddit users that posted to r/SuicideWatch, which may be more accurate measures of suicidality compared to whether or not users posted to r/SuicideWatch and thus could be used in future work.

Next, some of the aspects of the theories that our work focused on failed to be captured by variables in the Linguistic Inquiry and Word Count. For example, the state of narrowing of consciousness or cognitive deconstruction discussed by both Shneidman (1993) and Baumeister (1990) is an abstract concept and does not fit neatly into any of the dictionary categories of the LIWC. We attempted to use the all-or-nothing thinking variable (*allnone*) essentially as a proxy for this purpose, but it likely does not fully capture the essence of cognitive deconstruction. Future research may want to take advantage of the “build-your-own-dictionary” functionality of the LIWC, which allows researchers using the LIWC to define and score text using their own dictionary categories.

Additionally, we did not take as strict of an approach to model-building as we could have, which affected the predictive ability of our models; we could have performed a more rigorous model-selection process by adding or removing LIWC variables to test which combination resulted in the model with the best predictive power. The approach of the present study is relatively primitive compared to that of others in the field of natural language processing, and future research may benefit from more complex analyses and techniques, such as more sophisticated machine learning techniques.

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